

Quantitative Molecular Neurogenomics 2025

Module 6 - RNA-Seq Overview

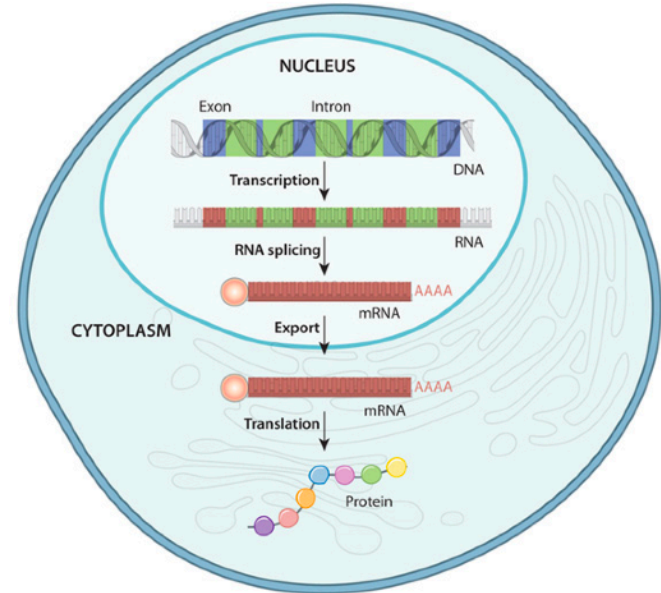
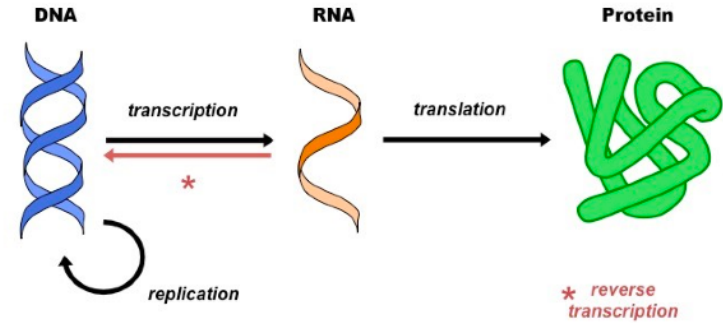
ME.440.825

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Gene Expression

- Information about cell type/state can be inferred from knowing what genes are **expressed** and when
- Differences in gene expression can highlight important changes and/or differentiating features
- mRNA quantification is a proxy measurement for (most) gene expression

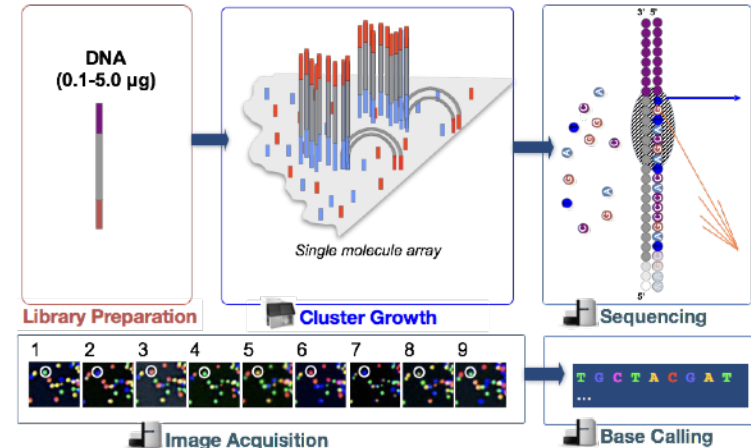
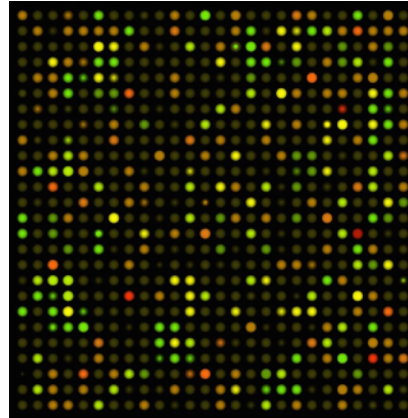
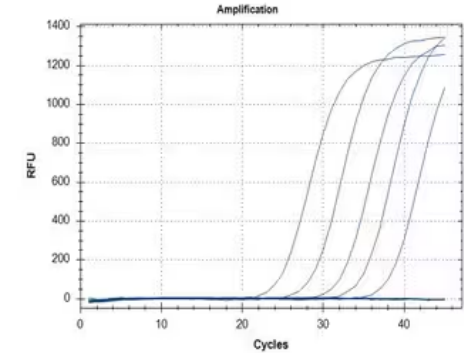
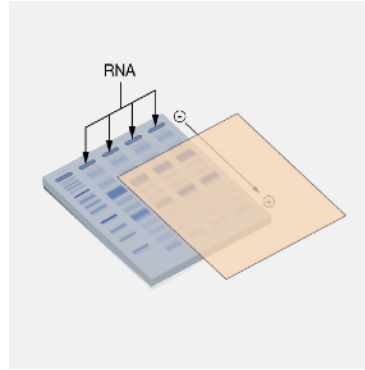


Why RNA?

- RNA->Protein rates are not captured
- Many more tools exist to manipulate/engineer/modify nucleic acids
- While not a ***perfect*** proxy, mRNA levels are more easily and accurately measurable *en masse* than proteins

Measuring Gene Expression

- Northern blot
- qRT-PCR
- Microarrays
- RNA-Seq



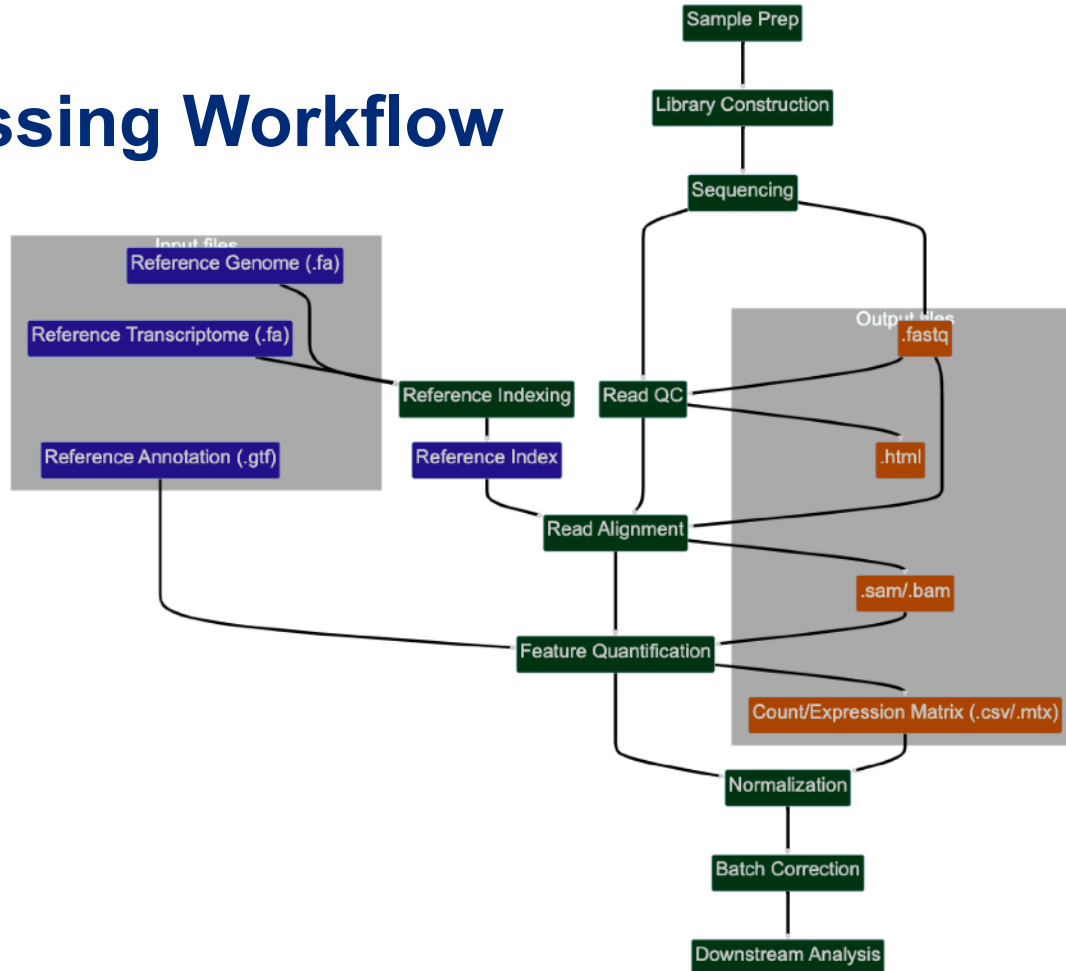
Why RNA-Seq?

- Cost-effective
- Unbiased
 - do not need to know anything about RNA content or sequence *a priori*
- Sensitive

What questions can be addressed with RNA-Seq?

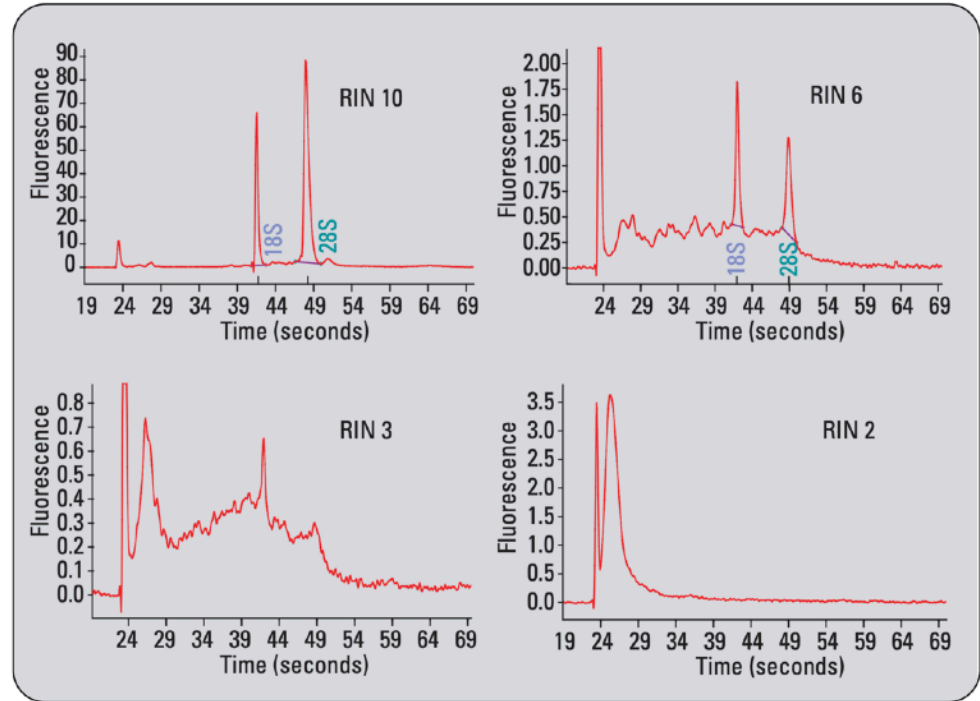
- Transcriptome assembly
- Novel transcript/isoform discovery
- ***Estimating*** gene expression
- Differential analysis
- Allele-specific expression
- Post-transcriptional modifications

RNA-seq Preprocessing Workflow



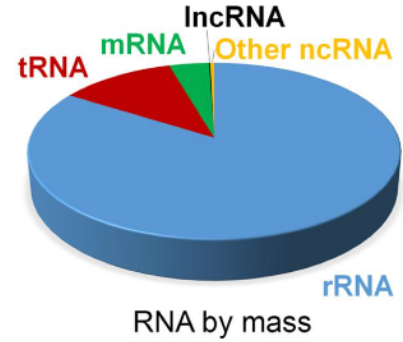
Sample Preparation

- Total RNA Extraction
 - Critical first step to obtain high-quality RNA
 - Purity and quality of extracted RNA are essential
- High-resolution gel or electropherogram essential to assess total RNA integrity
 - Quality measured by proxy of 18S & 28S rRNA integrity and ratio.
- Tissue/context and isolation method can affect RNA quality and yield

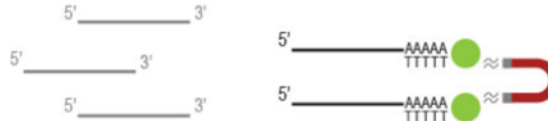


Poly(A) enrichment vs rRNA depletion

- Our target molecule is (usually) mRNAs and/or other PolII transcripts

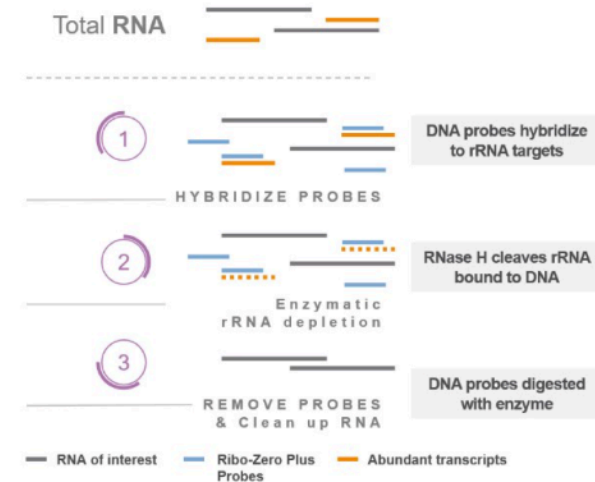


- Positive selection:
 - polyA-selection



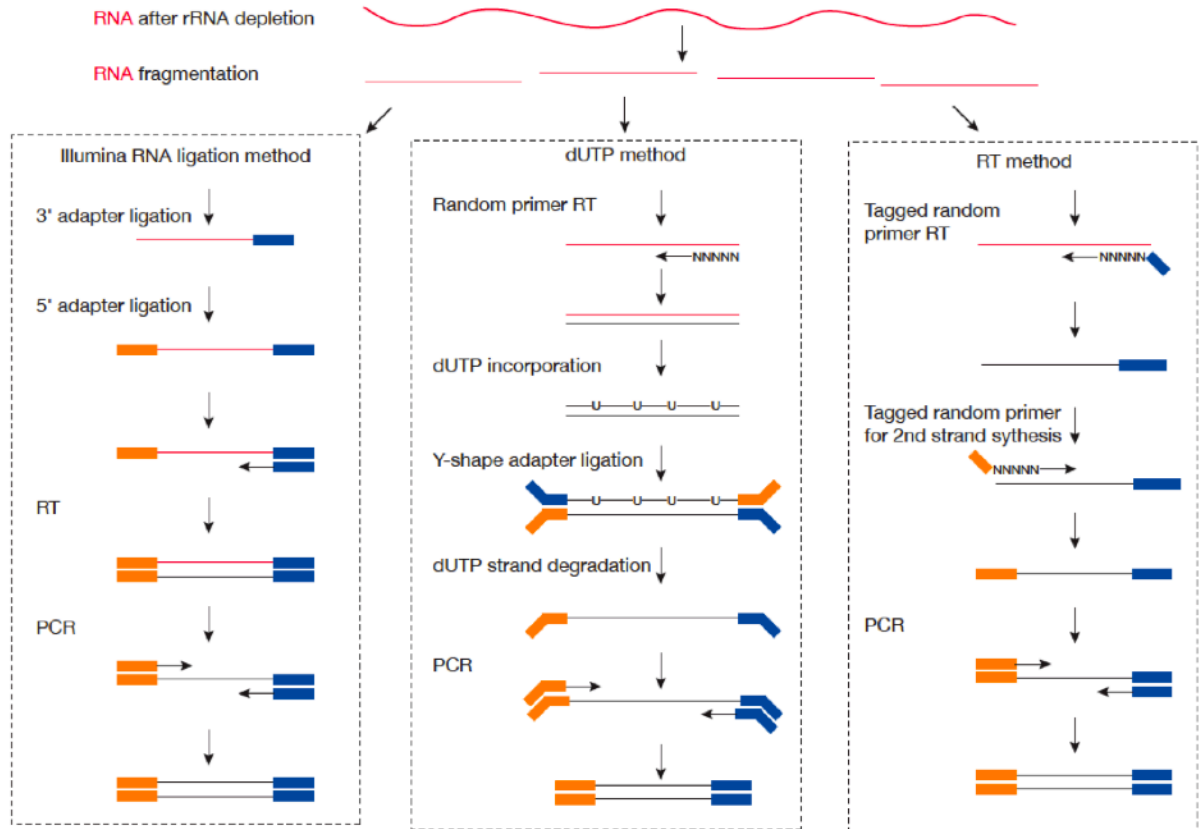
- Negative selection:
 - rRNA depletion

Ribo-Zero Plus: Enzymatic depletion



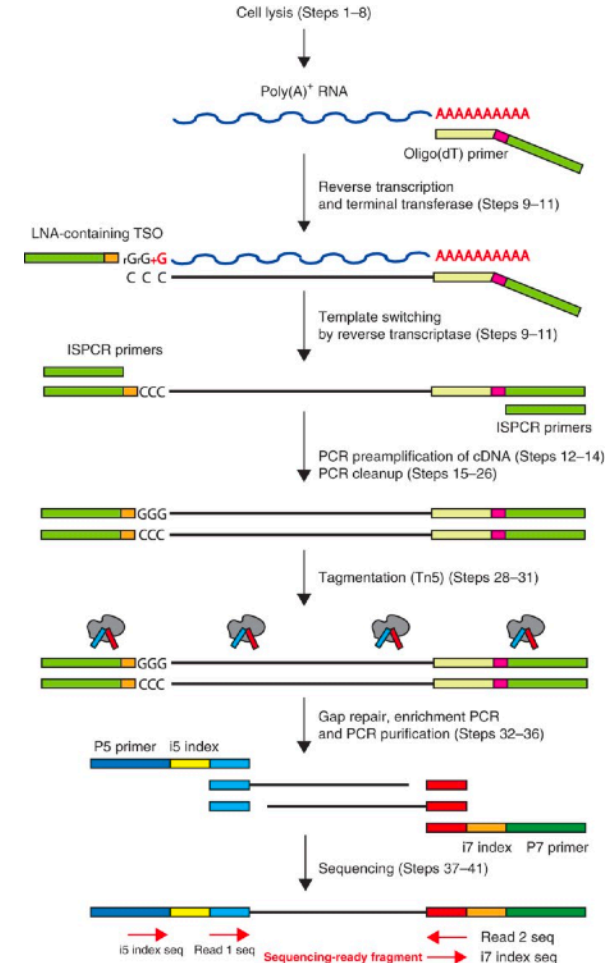
Library Preparation - Total RNA

- Useful for 'unbiased' assessment of RNA-species including non-polyA transcripts (e.g. lncRNAs)
- Pros:
 - Easy and high-efficiency preparations
 - Useful for finding non-PolIII transcripts
- Cons:
 - Depletion and/or reduction of rRNA is required beforehand



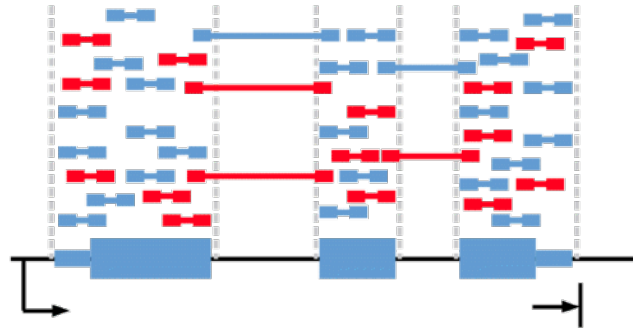
Library Preparation - mRNA

- Oligo(dT) primer targets poly(A) tail directly for RT
 - Reverse transcription converts RNA to complementary DNA (cDNA).
 - cDNA is fragmented and adapters are added for sequencing.
 - Library amplification via PCR.
-
- Pros:
 - Does not require rRNA depletion prior
 - Highly enriches for protein coding genes
 - Cons:
 - Can prime internally at homopolymer (A)_n sequences
 - Cannot detect RNAs w/o polyA tail

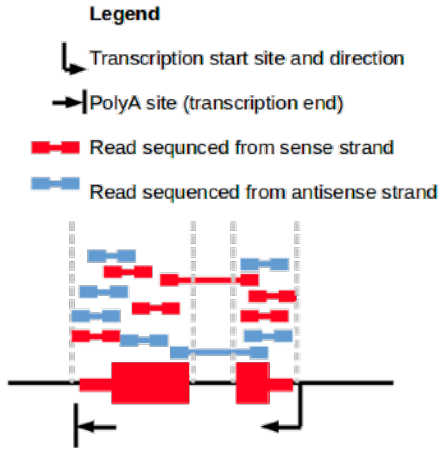


Strand-specific vs Unstranded library prep

A. Mapped reads from an unstranded library



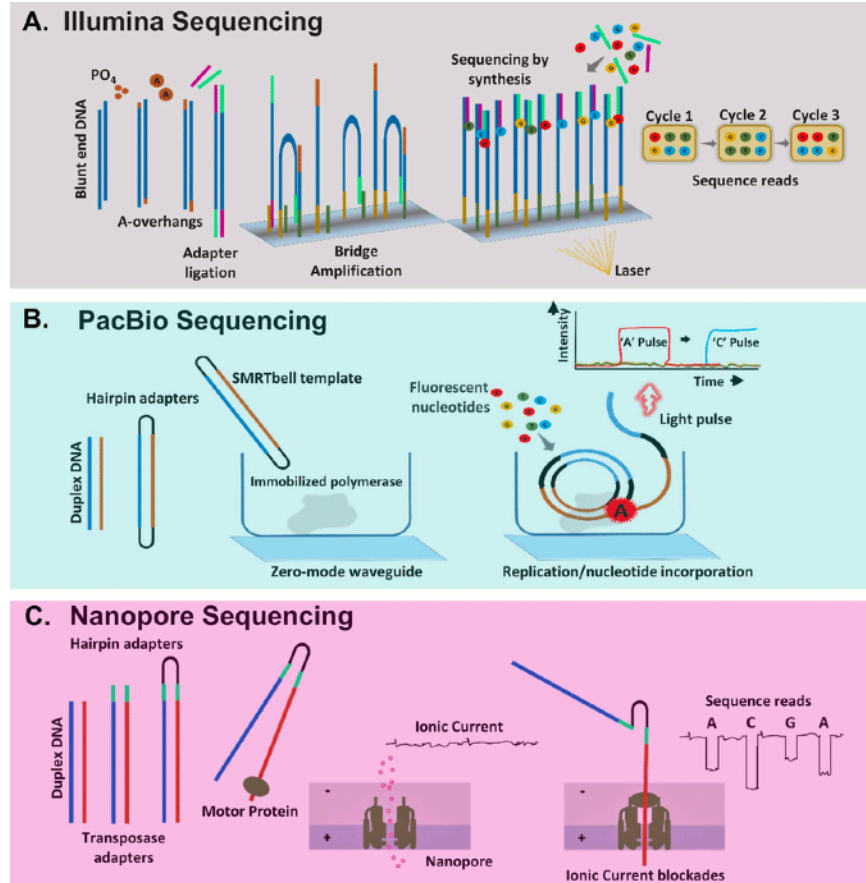
B. Mapped reads from a stranded library



- Older prep methods tended to produce ‘unstranded’ libraries
 - Less common these days
- Strand-specific libraries help clarify orientation and quantification
 - Especially for overlapping transcripts (e.g. lncRNAs)

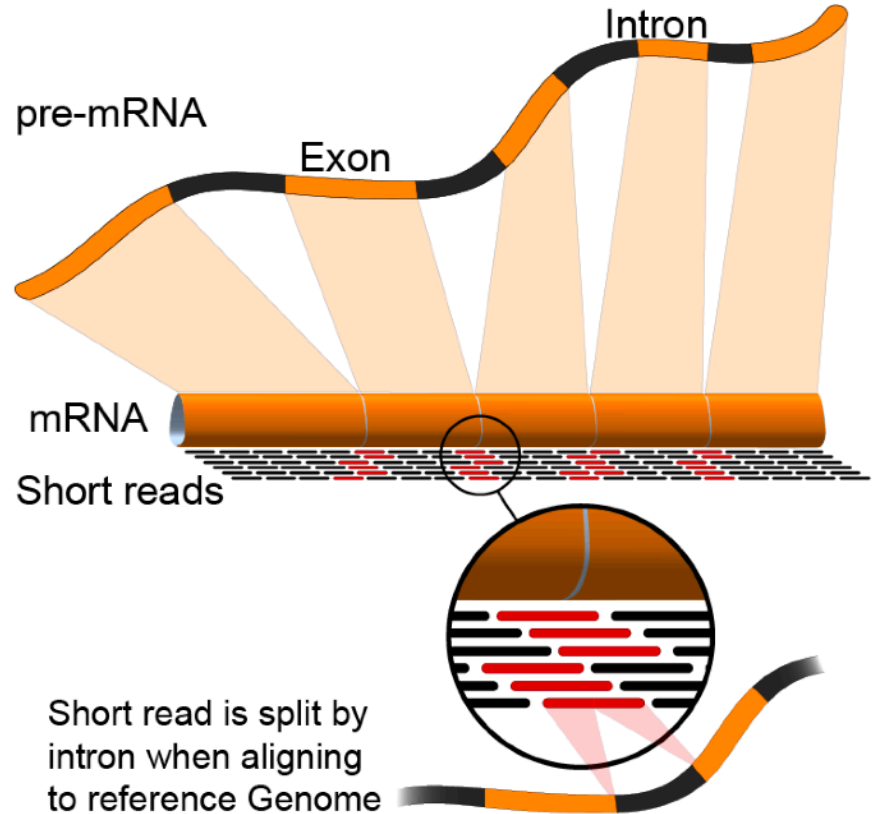
Short read vs Long read sequencing

- Often times, library prep is **very** similar for both methods.
- Comes down to choice of platform and question
- Illumina short read:
 - Produces between 100-600bp fragments
 - High accuracy (>99%)
 - Billions of fragments per sequencing run
 - Best for variant calling, differential expression, SNP detection.
 - Struggles with repetitive regions, needs a reference for assembly.
- Long read sequencing (e.g. Oxford Nanopore)
 - Several Kb to >2Mb reads, capturing whole transcripts
 - Lower accuracy (~90%)
 - Thousands to millions of reads per run
 - Ideal for de novo assembly, structural variants, splice variants, methylation detection.
 - Can **directly** measure RNA sequence



Alignment of RNA-seq fragments

- Reads derived from cDNA may contain exon-exon junctions
 - Longer & paired-end reads more useful in this context
- Require use of splice-aware aligner
- Junctions useful for identification & quantification of isoform-level expression

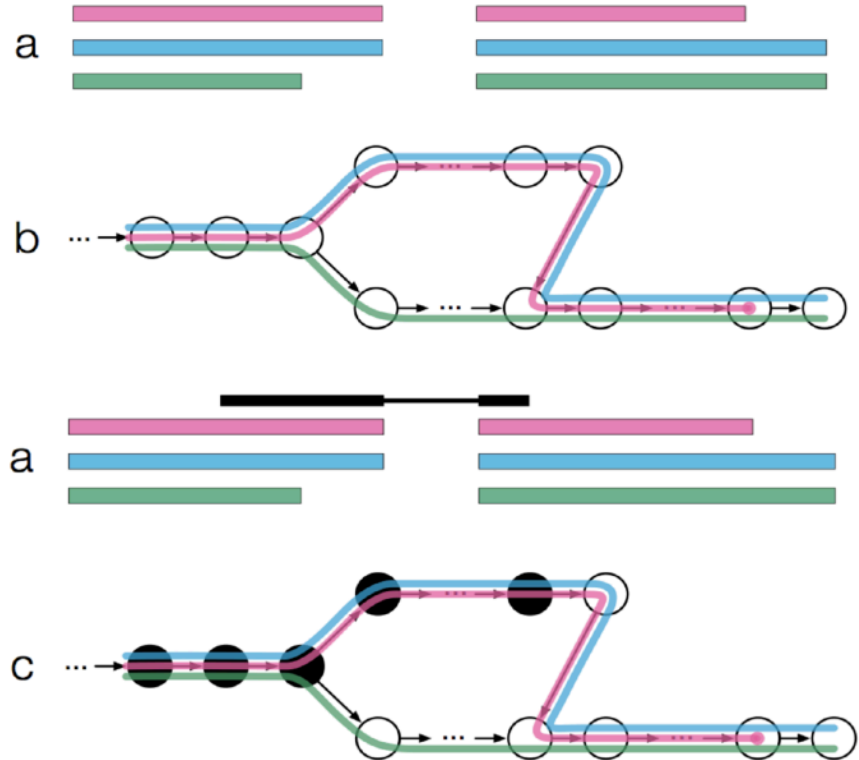


RNA: Alignment vs *Pseudoalignment*

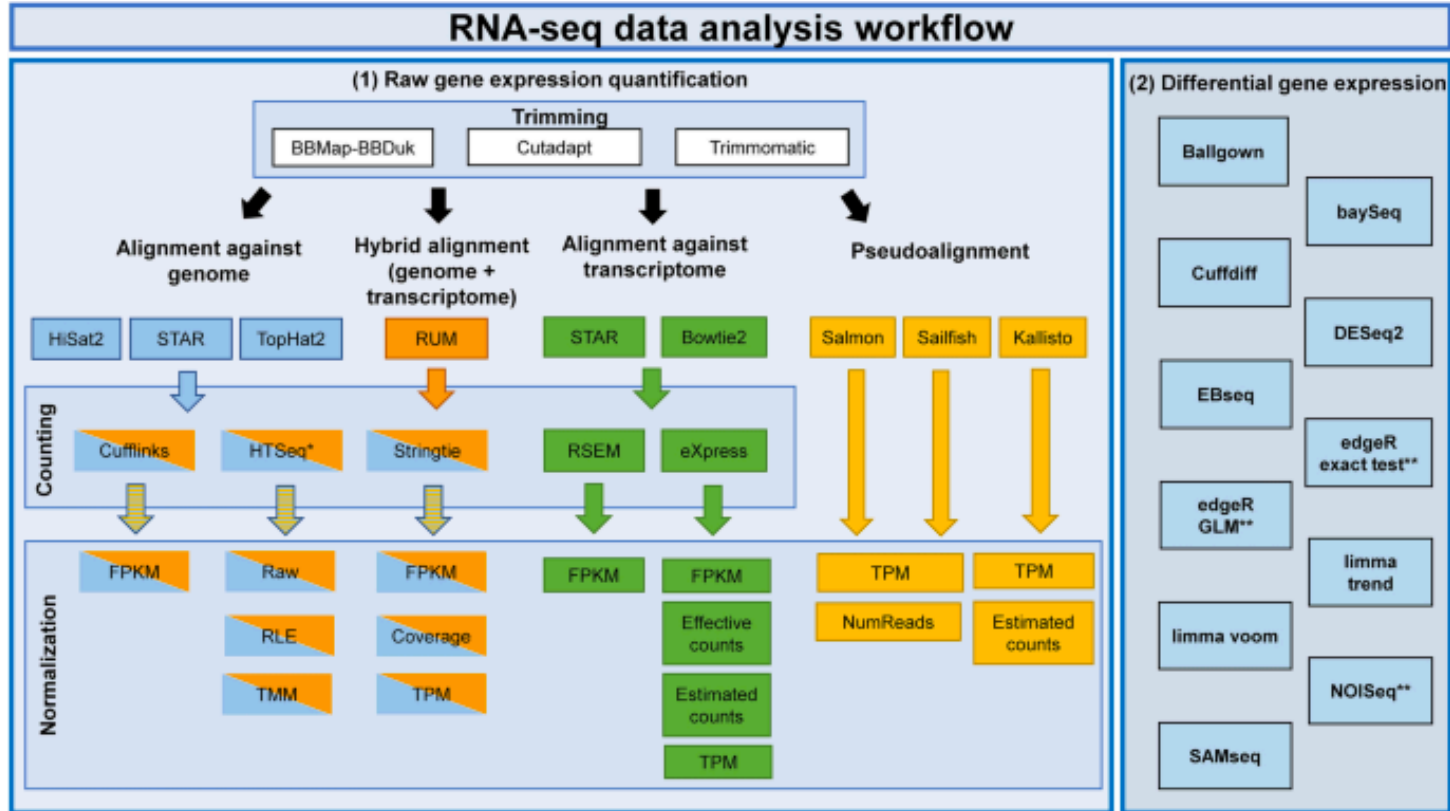
- Alignment
 - Mapping to reads to annotated reference genome
 - Variety of (splice-aware) alignment methods
 - Can be used for discovery of new transcripts
 - Speed depends on method but generally slower and more memory intensive
 - Alignment Tools
 - Tophat2
 - **HISAT2**
 - STAR
- Pseudoalignment
 - Mapping reads directly to reference transcriptome
 - K-mer-based equivalency class
 - No inference beyond reference transcriptome
 - Very fast
 - Speed allows for bootstrapping estimation of error in quantification
 - Pseudoalignment Tools:
 - **Kallisto**
 - Salmon

Pseudoalignment of RNA-Seq reads

- Different Goal:
 - ...*determine, for each read, not where in each transcript it aligns, but rather which transcripts it is compatible with...*
 - Useful for ‘counting-based’ problems
- Hashed index of Kmers that are ‘compatible’ with each reference transcript isoform
- Intersection of all k-compatibility classes for a read identifies the ‘compatible isoforms’ for a given read.

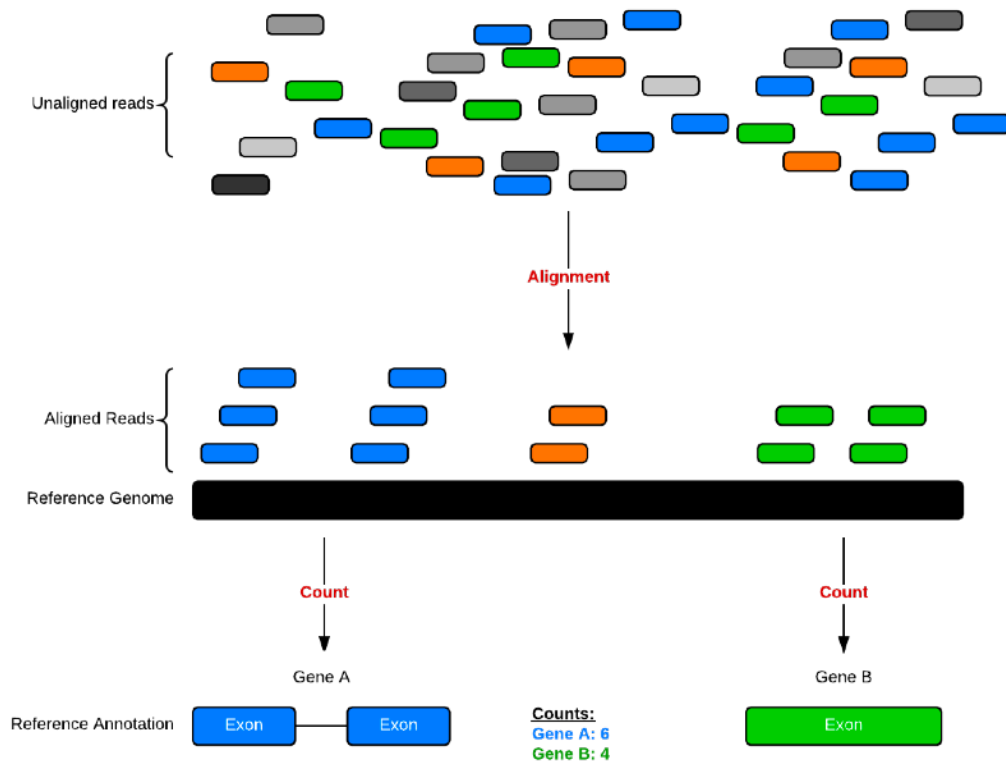


RNA: Alignment vs *Pseudoalignment*



Feature Quantification

- **Estimation** of the abundance of transcripts or genes in RNA-Seq data.
- **Objective:** Translate read alignments into meaningful measures of gene or transcript expression.
- Provides the foundational data for differential expression analysis, co-expression networks, and other downstream analyses.



Counting Methods

- Transcript-based
 - Counts reads uniquely aligned to individual transcripts (isoforms)
 - Kallisto or Salmon
 - Can be aggregated to gene-level post hoc
- Gene-based
 - Aggregates counts for all transcripts of a gene
 - featureCounts or HTSeq
 - Cannot be disambiguated to isoform-level
- The choice between transcript-based and gene-based quantification often depends on the research question.
- Transcript-based is more detailed but can be noisier, while gene-based provides a ‘cleaner’ summarized view.

Challenges in Feature Quantification

- Counts $\neq$ Expression
 - Gene length normalization required
 - Sequencing depth normalization required
- Ambiguous Mapping
 - Reads mapping to multiple regions/genes
 - Reads mapping to multiple isoforms
 - Solutions:
 - Drop multi-mapping reads from quantification
 - Distribute value across mapped features
- Biases
 - GC bias
 - PCR duplicates
 - Overall sequencing depth

Normalization Strategies / Units

- RPKM/FPKM

- Reads/Fragments per **K**ilobase of mRNA per **M**illion reads mapped
- Calculation:

$$\frac{n\text{Reads mapped/length in Kb}}{\text{Total reads in millions}}$$

- Values are **not** comparable across experiments

- TPM

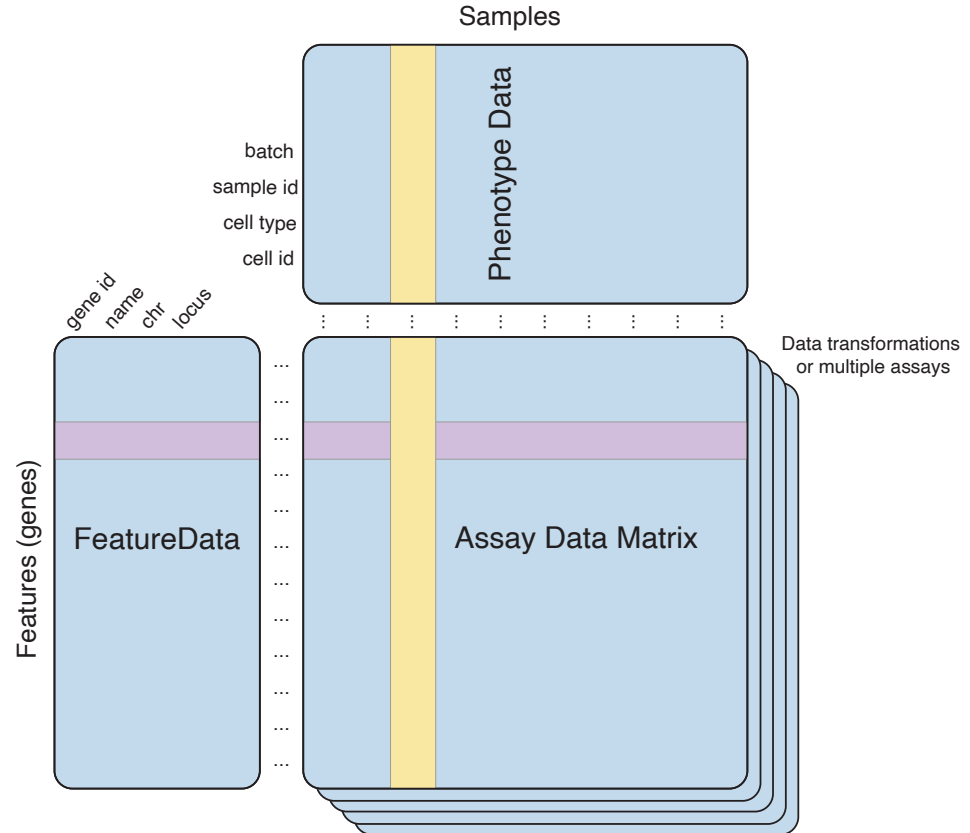
- Transcripts **P**er **M**illion
- Calculation:
 - First normalize by RPKM/FPKM
 - Divide by the sum of all normalized read counts and multiply by 1 million.
- Values are now comparable across experiments

Raw count normalization as part of DE

- Several differential expression tools instead directly model the ***raw read counts*** using a negative binomial distribution.
 - Input should ***not*** be pre-normalized as it is handled internally
 - e.g. DESeq2, edgeR
- These tools use scaling factors derived from the data to adjust for library size differences.
 - For instance, DESeq2 uses the median-of-ratios method.

Aggregation of sample counts to matrix

- Each RNA-Seq sample produces count data for genes/transcripts.
- Aggregating these counts creates a comprehensive expression matrix.
 - Concatenating vectors together provides the foundational matrix for all downstream analyses
- Ensure consistent naming/annotation across samples.
 - Merging/reordering genes is a common error during this step
 - Be sure to validate/check often
- Merge sample metadata and retrieve feature annotation at this stage as well.



Tools and Resources

Neuroscience

Hipposeq: a comprehensive RNA-seq database of gene expression in hippocampal principal neurons

Mark S Cembrowski, Lihua Wang, Ken Sugino, Brenda C Shields, Nelson Spruston 

Janelia Research Campus, Howard Hughes Medical Institute, United States

Apr 26, 2016 · <https://doi.org/10.7554/eLife.14997>   Download Cite Comment

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